ForgeSlicer

Comprehensive Manual QA Test Plan

Version 1.0 · June 27, 2026

Environment: **https://orca-cad-slice.preview.emergentagent.com**

ForgeSlicer is a browser-based 3D CAD + Slicer that fuses a TinkerCAD-style primitive editor, Shapr3D-style reverse engineering, and an OrcaSlicer-style slicing handoff into a single workflow. This document is the comprehensive manual QA test plan that walks a tester through every user-facing surface — primitives, Boolean operations, importers, RANSAC reverse engineering, AI / voice workflows, the community gallery, Learn lessons, Trust pages, account and email flows, and slicer handoff.

Each test case has a stable ID (so regressions can be referenced across releases), the click-by-click steps a tester should follow, the expected result, and a priority. Run P0 cases on every release, P1 cases weekly, and P2 cases when a related area changes.

Test environment: **https://orca-cad-slice.preview.emergentagent.com**. Recommended browsers: latest Chrome, Firefox, and Safari. Use Chromium-based browsers for WebGPU-dependent paths.

P0 Run every release	**P1** Run weekly	**P2** Run on related changes

Contents

1. Onboarding & First-Run

Beginner Starters

ID	Test Case	Steps	Expected Result	Pri
ONB-01	Beginner Starters carousel renders 12 templates	1. Open / in a private window. 2. Click 'Start designing'. 3. Observe the Beginner Starters strip.	12 starter cards render with thumbnails, titles, and 'Open in editor' CTAs. No console errors.	**P0**
ONB-02	Opening a starter populates the canvas	1. Click any starter (e.g. 'Phone Stand'). 2. Wait for the editor to mount.	Canvas loads the starter geometry, scene tree shows the template's nodes, undo history is empty.	**P0**
ONB-03	Skip onboarding banner is dismissible & non-blocking	1. Land on /. 2. Locate the bottom-right announcement banner. 3. Click the X.	Banner closes, never reappears in the same session, no fullscreen modal interrupts work.	**P1**

Auth (email + Google)

ID	Test Case	Steps	Expected Result	Pri
AUTH-01	Email + password signup → email verification	1. /signup. 2. Use a fresh address. 3. Submit. 4. Check inbox.	Account row created, verification email arrives via Resend, link sets `verified=true`.	**P0**
AUTH-02	Magic link sign-in	1. /signin. 2. Click 'Send magic link'. 3. Click link in email.	Single-use link signs the user in within 15 minutes; reusing it returns an expired error.	**P0**
AUTH-03	Password reset	1. /signin → 'Forgot password'. 2. Submit email. 3. Open link. 4. Set new password.	Reset link valid 60 min, signs user out of all sessions on success.	**P0**
AUTH-04	Emergent-managed Google sign-in	1. /signin → 'Continue with Google'. 2. Approve the consent screen.	User is created or matched on email, profile photo + name populate.	**P0**

2. Primitive Editor & Boolean Operations

Primitives

ID	Test Case	Steps	Expected Result	Pri
PRIM-01	Insert cube / sphere / cylinder / cone / torus / triangle	1. Editor → Primitives palette. 2. Click each shape in turn.	Each primitive spawns at origin with sensible defaults; scene tree updates; gizmo attaches.	**P0**
PRIM-02	Edit primitive parameters (dims, segments)	1. Select a cylinder. 2. Open the Parameters panel. 3. Change radius, height, segments.	Mesh regenerates live, no flicker, history step recorded.	**P0**
PRIM-03	Triangle primitive — equilateral default	1. Insert triangle. 2. Inspect dimensions panel.	Equilateral by default; PRD note: configurable base/height/angles is upcoming P1.	**P1**
PRIM-04	Transform: move / rotate / scale via gizmo	1. Select primitive. 2. Press G / R / S (or click toolbar). 3. Drag axes.	Snapping to 1 mm / 15° works with Shift; numeric panel reflects gizmo state.	**P0**

Boolean Operations (Manifold-3D WASM)

ID	Test Case	Steps	Expected Result	Pri
BOOL-01	Union two overlapping cubes	1. Insert 2 cubes overlapping. 2. Multi-select. 3. Toolbar → Union.	Result is a single watertight mesh; volume ≈ sum minus overlap; history step recorded.	**P0**
BOOL-02	Subtract cylinder from cube (hole)	1. Insert cube and cylinder passing through it. 2. Select cube then cylinder. 3. Subtract.	Hole punched cleanly; resulting mesh is manifold; no inverted normals.	**P0**
BOOL-03	Intersect cube ∩ sphere	1. Place a sphere overlapping a cube. 2. Intersect.	Yields the cube-corner-cut-by-sphere lens; mesh is closed; saves to project.	**P0**
BOOL-04	Boolean on non-manifold input shows healing toast	1. Import a known non-manifold STL. 2. Attempt Boolean.	Healing pass runs; if unrecoverable, user sees clear error toast instead of a crash.	**P1**

3. Importers (STL / OBJ / 3MF / SVG / ZIP)

Mesh imports

ID	Test Case	Steps	Expected Result	Pri
IMP-01	STL drag-drop	1. Drag any STL onto the canvas.	Mesh appears centered, axis-aligned bbox shown; units default mm.	**P0**
IMP-02	OBJ with material groups	1. Import an OBJ that has multiple groups.	Groups become separate scene-tree nodes preserving names.	**P0**
IMP-03	3MF multi-object project	1. Import a 3MF authored in PrusaSlicer or Bambu Studio.	All objects retained with their original transforms; print-settings warning if present.	**P0**
IMP-04	SVG extrude	1. Drop an SVG. 2. Set extrude depth in the modal.	2D paths become a solid mesh; bezier curves sampled smoothly.	**P1**
IMP-05	ZIP archive (mixed assets)	1. Drop a ZIP containing STL + 3MF + SVG.	Importer enumerates contents, asks the user to choose, imports the selection.	**P1**

4. Reverse Engineering (RANSAC)

Dialog & detection (Phase 1–3 shipped)

ID	Test Case	Steps	Expected Result	Pri
RE-01	Open Reverse Engineer dialog from a selected mesh	1. Select an imported mesh. 2. Toolbar → 'Reverse engineer'.	Modal opens, sensitivity controls visible, detection runs and lists primitives.	**P0**
RE-02	RANSAC detects a box+cylinder composite	1. Import the 'bracket.stl' sample. 2. Run detection.	Result lists at least 1 box and 1 cylinder with sane dimensions (±5%).	**P0**
RE-03	Sensitivity slider (Phase 4, upcoming)	1. Drag the sensitivity slider. 2. Re-run detection.	Higher sensitivity yields more primitives, lower yields fewer; no infinite spinner.	**P1**
RE-04	Replace with primitives button (Phase 5, upcoming)	1. Click 'Replace with primitives'.	Original mesh hidden, parametric Three.js primitives spawned at detected transforms; undo restores mesh.	**P0**

5. AI & Voice Workflows

Conversational AI

ID	Test Case	Steps	Expected Result	Pri
AI-01	Voice command: 'make a 20 mm cube'	1. Click mic. 2. Speak the command.	Whisper transcribes; GPT-5.2 returns JSON intent; cube of side 20 mm appears.	**P0**
AI-02	Meshy AI: text → 3D	1. AI panel → 'Text to 3D'. 2. 'low-poly rocket'. 3. Submit.	Job polls Meshy; result mesh imports into the scene when ready; failure surfaces friendly toast.	**P0**
AI-03	Meshy AI: image → 3D	1. Upload an image to the AI panel. 2. Generate.	Returns watertight mesh; preview thumbnail accurate.	**P1**
AI-04	Meshy attribution copy	1. Open AI panel.	Footer explicitly states 'Powered by Meshy.ai — third-party service'.	**P1**

Text on Surface

ID	Test Case	Steps	Expected Result	Pri
TXT-01	Text-on-surface MVP	1. Select a flat face. 2. Tools → Text on surface. 3. Type 'Hi'. 4. Apply.	Helvetiker text mesh extrudes onto the face; aligned to face normal; Boolean-union onto host.	**P0**

6. Measurement, Snapping & Grid

Measurement

ID	Test Case	Steps	Expected Result	Pri
MEAS-01	Linear measurement between two vertices	1. Tool → Measure. 2. Click 2 vertices.	Distance label shown in mm; persists until cleared; respects unit toggle.	**P1**
MEAS-02	Grid + snapping toggles	1. View menu → Toggle grid / Toggle snapping.	Grid renders at 10 mm; snapping locks transforms to 1 mm; toggles persist across reloads.	**P1**

7. Community Gallery (v2)

Browse & filter

ID	Test Case	Steps	Expected Result	Pri
GAL-01	Public gallery loads with taxonomy	1. Open /gallery.	Categories sidebar visible (Functional, Decorative, Educational…), pagination works, featured creators row populated.	**P0**
GAL-02	Filter by tag + category	1. Select category 'Functional' + tag 'organizer'.	Result set narrows; URL reflects state; deep link reproducible.	**P0**
GAL-03	Visibility: private item hidden in public gallery	1. As user A publish private item. 2. Sign out. 3. Browse gallery.	Private item NOT listed publicly but still appears in A's profile.	**P0**
GAL-04	Contributor Lifetime celebration email	1. As an account, cross 100 components + 20 designs.	Resend dispatches the celebration email; badge appears on profile.	**P1**

8. Learn Section

Lessons

ID	Test Case	Steps	Expected Result	Pri
LRN-01	/learn lists 8 lessons	1. Open /learn.	8 lesson cards render with duration + difficulty tags.	**P1**
LRN-02	Lesson detail page deep-links	1. Click a lesson. 2. Copy URL. 3. Open in incognito.	Lesson loads directly, meta tags + structured data correct.	**P1**

9. Trust, Privacy & Changelog

Trust pages

ID	Test Case	Steps	Expected Result	Pri
TRUST-01	/trust renders policy summary	1. Open /trust.	Sections: data handling, third-party services (Meshy, OpenAI, Resend), security posture.	**P1**
TRUST-02	/privacy + /changelog accessible from footer	1. Scroll to footer on any page.	Both links present and resolve to readable pages.	**P1**

10. Slicer Handoff

Multi-object 3MF export → OrcaSlicer

ID	Test Case	Steps	Expected Result	Pri
SLI-01	Send to OrcaSlicer (desktop)	1. Build a scene with 3 objects. 2. Toolbar → 'Send to slicer' → Orca.	3MF downloads with all objects, transforms, and modifier volumes; opens in Orca without warnings.	**P0**
SLI-02	PrusaSlicer / Bambu Studio variants	1. Repeat with Prusa and Bambu targets.	Each slicer opens the 3MF cleanly; vendor-specific profile metadata respected when set.	**P1**
SLI-03	Watertight check before export	1. Force a non-manifold mesh. 2. Attempt export.	Modal warns and offers auto-heal; export blocked until user accepts or cancels.	**P0**

11. SEO & Marketing Surfaces

SEO landings

ID	Test Case	Steps	Expected Result	Pri
SEO-01	Targeted landings render unique meta	1. Visit each of the 8 SEO pages from /seo (or sitemap).	Each page has unique <title>, <meta description>, OG tags, JSON-LD.	**P2**
SEO-02	sitemap.xml + robots.txt valid	1. curl /sitemap.xml and /robots.txt.	XML validates, robots references the sitemap, no 5xx.	**P2**

12. Cross-Cutting Quality Bars

Performance & accessibility

ID	Test Case	Steps	Expected Result	Pri
QA-01	Cold load TTI < 4 s on cable	1. Lighthouse on /editor.	Performance score ≥ 80, no render-blocking JS warnings.	**P1**
QA-02	Keyboard navigation across modals	1. Tab through Reverse Engineer dialog.	Focus trap honored, ESC closes, no focus-loss bugs.	**P1**
QA-03	Mobile responsive sanity	1. Open / on a 390 px-wide viewport.	Hero, gallery, learn render without horizontal scroll; editor shows a gentle 'desktop recommended' note.	**P2**

Sign-off

When all P0 cases pass and no P1 regressions are open, this build is ready to ship. Record the tester, date, browser, and any deviations below before archiving.

Tester	
Date	
Browser / OS	
Build / commit	
Notes	